

Sexual selection's role in the persistence of polymorphism in an aposematic signal

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Abstract

The persistence of polymorphisms in aposematic species remains one of the most interesting paradoxes in evolutionary biology because aposematism theory suggests that polymorphisms should be unstable over time. We offer an explanation for the persistence of aposematic polymorphisms that considers not only the role of natural selection but also the role of sexual selection. While predation and mate choice generally act to erode signal variation, intraspecific competition may facilitate polymorphisms by reducing mate competition for males bearing the rarer warning signal. We tested this hypothesis in a population of the strawberry poison frog (*Oophaga pumilio*), where red (common) and yellow (rare) morphs coexist using (1) a 10-year capture–mark–recapture experiment to study natural selection, (2) a territorial intrusion experiment, and (3) previously published mate choice experiments to study sexual selection. We found that rare yellow males suffer less aggression from male conspecifics, suggesting negative frequency dependent selection. Moreover, the more common and choosier red females have lower apparent survival than their less choosy yellow counterparts, suggesting that survival may be better explained by costs of exercising mate choice rather than by predation. Our work highlights the importance of considering multiple sources of selection in explaining the paradoxical persistence of aposematic polymorphisms.

Keywords: aposematism, polymorphism, sexual selection, predation, frequency dependent selection

Introduction

Aposematism, a strategy by which animals advertise their unprofitability to predators through warning signals (Poulton, 1890), has driven the evolution of a wide range of conspicuous and distinctive traits in nature (Caro & Ruxton, 2019; Mappes et al., 2005; Rojas et al., 2015). While warning signals can be of various modalities such as chemical or acoustic cues (Hristov & Conner, 2005; Weldon, 2013), coloration is a common, easily observed, and well-studied type of warning signal. Aposematism theory posits that predators must learn to associate a signal with a secondary defense, and such learning is most efficient when the warning signal is consistent within a population (Rojas et al., 2015). As such, aposematic traits are under positive frequency dependent selection, and polymorphic warning signals (i.e., the occurrence of multiple signal variants within a population) are expected to be evolutionarily unstable (Mallet & Joron, 1999; Ponshe & Endler, 2022; Richards-Zawacki et al., 2013; Rojas et al., 2015). Yet, warning signal variation and polymorphism within aposematic populations is not uncommon in nature. For example, it has been documented in frogs (Richards-Zawacki et al., 2013; Rojas & Endler,

2013), moths (Henze et al., 2018), and beetles (Brakefield, 1985; Doktorová et al., 2019; Roggero et al., 2024). Interestingly, this phenomenon is not restricted to populations of the same species but can be observed among co-occurring aposematic mimic species (e.g., butterflies; Mallet & Joron, 1999) (Figure 1). We argue that considering the influence of both forms of sexual selection (i.e., mate choice and intraspecific competition) in addition to predation (the presumed source of natural selection) on warning signals can help to explain this long-held paradox in evolutionary biology.

While aposematism is usually considered an evolutionary product of natural selection, recent empirical and theoretical work suggests roles for sexual selection in the evolution and maintenance of warning color polymorphisms (Briolat et al., 2019; Ponshe & Endler, 2022; Rojas et al., 2018). Most theoretical studies consider the potential for the less protected aposematic phenotype (i.e., the rarer one) to be more attractive to the opposite sex, a tradeoff that could stabilize the coexistence of multiple phenotypes that have the same total fitness (Nokelainen et al., 2011; Ponshe & Endler, 2022). However, empirical evidence does

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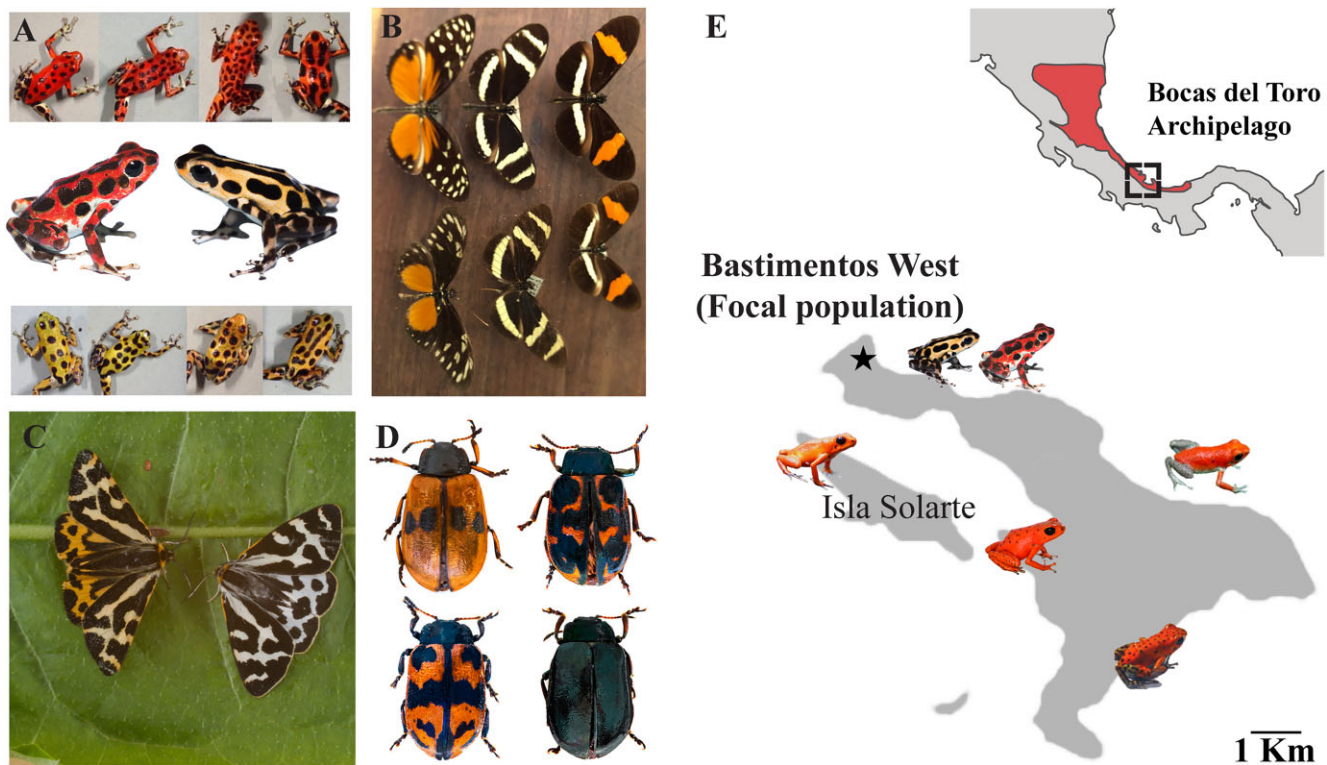


Figure 1. (A) Aposematic color variation of the strawberry poison frog, *Oophaga pumilio*, from the Bastimentos West population investigated in our study, (B) Aposematic color variation seen among different species of *Heliconius* butterflies found in the Osa Peninsula of Costa Rica, (C) individuals of the tiger moth, *Arctia plantaginis*, in Finland, and (D) individuals of the leaf beetle *Chrysomela lapponica* found across Europe and Russia. In (E), the larger bottom map shows the aposematic *O. pumilio* phenotypes found on Bastimentos and nearby Solarte Island. The red area in the upper right map shows the entire distribution of *O. pumilio* on a map of southern Central America. Photo/artwork credits: (A and E) Corinne Richards-Zawacki and Vincent Prémel (B) Andrés Vega, (C) Samuel Waldron, (D) Vitali Zverev, and previously published images in Doktorová et al. (2019) obtained with permission.

not always match this theoretical prediction (e.g., Gordon et al., 2015; Roggero et al., 2024), suggesting that additional processes might be required for polymorphism maintenance. Intersexual selection, where individuals of one sex (often females) choose mates based on specific traits exhibited by individuals of the other sex, and intrasexual selection, where individuals of the same sex (often males) compete for mates through physical aggression or other forms of competition, are the two main mechanisms of sexual selection (Andersson, 1994). While seldom considered in the context of aposematism, competition mediated by warning signal variation could promote the maintenance of aposematic polymorphisms by reducing the competition rare morphs face in securing territories or access to members of the opposite sex. If males with the more common warning signal bias their aggression toward other males bearing their own phenotype, then males of the rarer phenotype gain an advantage via negative frequency dependent selection (Mikami et al., 2004; Seehausen & Schluter, 2004; Yang, Servedio et al., 2019). While the idea that intraspecific competition can stabilize trait polymorphisms is not new (Dijkstra et al., 2005; Pryke & Griffith, 2005; Zamudio & Sinervo, 2000), it has thus far mostly been investigated in nonaposematic animals.

The strawberry poison frog, *O. pumilio*, shows evidence of recent and rapid divergence in aposematic coloration both among and within populations (Brown et al., 2010; Maan & Cummings, 2012; Saporito et al., 2007; Summers et al., 2003). The bright warning coloration seen in this species is also used by females in mate choice (Maan & Cummings,

2008; Reynolds & Fitzpatrick, 2007; Richards-Zawacki & Cummings, 2011; Richards-Zawacki et al., 2012; Summers et al., 1999; Yang et al., 2016; Yang, Servedio et al., 2019) and by males in rival recognition (Yang et al., 2018, 2019). Interestingly, color biases in both female mate preference and male aggression are known to be shaped by learning (i.e., maternal imprinting) (Yang et al., 2019). Most *O. pumilio* populations are monomorphic in color, but a few are polymorphic (Richards-Zawacki & Cummings, 2011; Summers et al., 2003; Yang et al., 2016).

One well-studied polymorphic population is found at the northwest tip of Bastimentos Island in the Bocas del Toro Archipelago of Panama, where conspicuous red- and yellow-colored frogs co-occur (Richards-Zawacki et al., 2013). Although both morphs appear to be equally unpalatable based on skin alkaloid profiles (Saporito et al., 2006) the red morph is more common within the polymorphic population, and red is the most common color among neighboring monomorphic populations as well (Richards-Zawacki & Cummings, 2011; Richards-Zawacki et al., 2013) (Figure 1E). Based on traditional aposematism theory, the regional predominance of red as an aposematic signal should facilitate predators' learning to avoid the red morph more easily than the yellow one. However, a previous study failed to detect differential predation among the two color morphs (Richards-Zawacki et al., 2013). For mate choice, both red and yellow females from this population show color-assortative *mate preferences* in laboratory choice assays (Richards-Zawacki & Cummings, 2011), but reconstruction

of a wild pedigree revealed assortative *mating* only in red females, whereas yellow females appear to mate randomly with respect to male color (Richards-Zawacki et al., 2012). Such mating patterns should lead to the continued increase and eventual fixation of the preferred and more common red color. The persistence of this polymorphism suggests the existence of a fitness advantage for the rare yellow morph, and neither natural selection nor intersexual selection appear to generate this. We hypothesized that reduced competition due to biased male–male aggression, a negative frequency dependent form of intrasexual selection (Yang et al., 2019) creates this advantage for the rarer of the two warning signals.

To test this hypothesis, we built on our previous work on the polymorphic Bastimentos West *O. pumilio* population to further characterize the roles of natural and sexual selection in maintaining an apparently stable aposematic polymorphism. We considered results from prior work on female mate choice and present new data from a combination of field territorial intrusion assays and capture–mark–recapture surveys done over a 10-year period to test two main predictions: (1) that the more common red morph has higher survival rates than the yellow morph, as would be expected if predation was exerting positive frequency dependent natural selection on the aposematic signal, and (2) that males bias their aggression toward their own phenotypes, which could stabilize the aposematic polymorphism by providing more territories and greater access to mating opportunities for the rare yellow morph.

Methods

Study site

We conducted a study of *O. pumilio* on Bastimentos Island in the Bocas del Toro Archipelago of Panama (Figure 1). The study area was ~7,500 m² of predominantly secondary-growth forest. Within the study area, two color morphs, distinguishable by human eye as well as in the visual system of *O. pumilio* and their potential predators (Richards-Zawacki et al., 2013), can be found coexisting and interacting with one another. Frogs can be categorized as either red (red to orange dorsal and limb coloration), or yellow (yellow to occasionally a minty-green dorsal and limb coloration; Figure 1A). Coloration seems to follow a simple mendelian inheritance pattern in which red is dominant while yellow is recessive (Richards-Zawacki et al., 2012). Moreover, a recent study suggests that yellow coloration may be the result of a mutation that disrupts the metabolic pathway for yellow-to-red carotenoid conversion (Aguilar-Gómez et al., unpublished). Both morphs have black spots or blotches on their dorsum and limbs and a pure white venter.

Capture–mark–recapture surveys

Oophaga pumilio is diurnal and active throughout the year. We surveyed the study area 15 times between 2007 and 2017 (Figure 2B and D). Each survey was performed by a team of three to five people during one to three survey days within a 4-day period. During each survey, we captured frogs by hand and transported them to the Bocas del Toro Research Station at the Smithsonian Tropical Research Institute, where we took standardized photographs using a digital camera and measured body size (snout–vent–length, SVL) using dial calipers. We housed captured frogs in small plastic contain-

ers with vented lids and either moist paper towels or natural leaf litter as a substrate, and fed them daily with termites and fruit flies while we took measurements from each individual. We released all frogs back to the study area at the end of each survey, after a maximum of 4 days in captivity.

We used photographs of the dorsum and venter of individual frogs as noninvasive “marks” black dorsal spotting patterns, which are highly variable, allow for accurate identification of individuals from photographs (~0.3% misidentification rate; Richards-Zawacki et al., 2012; Figure 1A). To detect recaptures, we divided the photographs up into groups by frog sex using the presence of a darkened throat patch on the ventral photo to distinguish males (~3.6% error rate for sex determination; Richards-Zawacki et al., 2012). We classified individuals as adults based on their body size (i.e., ≥18 mm in SVL). Frogs smaller than 18 mm SVL (range: 16.3–17.9 mm, *N* = 16) were occasionally found to have dark throat patches and these were classified as male. We manually compared the photos of all captured individuals in each group (male, female, and juvenile) with all prior photos in the same group after each survey. All male and female photos were also compared with all prior juvenile frogs as frogs may have reached maturity between survey periods. This allowed us to classify the captured individuals in each survey as recaptures (match to a photo from a prior survey) or new individuals (no pattern match).

Modeling the effect of polymorphism on survival

We used the open population Cormack–Jolly–Seber (CJS) model implemented in the R package *RMark* (Laake, 2013) to estimate the apparent survival (Φ) and probability of detection (p) of adult individuals of *O. pumilio* and test the hypothesis that these parameters differ among the red (red to orange individuals) and yellow (yellow to minty-green individuals) color morphs (as delimited in Richards-Zawacki et al., 2013). CJS models assume that marked individuals at time i have the same probability of survival to time $i + 1$ and the same probability of recapture at time $i + 1$. Given that males and females differ in reproductive behavior and habitat use (e.g., calling, territoriality, and egg care in males vs. tadpole transport and provisioning in females, reviewed in Dugas, 2018) in ways that could influence both survival and recaptures during the survey, and since we were primarily interested in testing for differences between the colors rather than among sexes, we analyzed males and females separately.

For each sex, we built a total of 16 alternative models where survival and detection were either held constant, or allowed to vary by color morph, sampling period, or the interaction between color and sampling period. Because sampling periods were not equidistant (Figure 2B and D), the models were calibrated by specifying the time between sampling periods in *RMark*. To confirm the fit of the models to our data, we performed a goodness of fit test and estimated the median variance inflation factor ($\hat{c}$) of the full model (i.e., when both survival and detection are affected by both color and sampling period) using the software *MARK* following (Cooch & White, 2019). After confirming the fit of the models to our data, we proceeded to select the model that best fit our data using the Akaike information criterion (AIC) approach in *RMark*. We corrected the values of the AIC of every model using the $\hat{c}$ value estimated for the full model (i.e., we used Akaike’s small-sample-size-corrected quasi-likelihood infor-

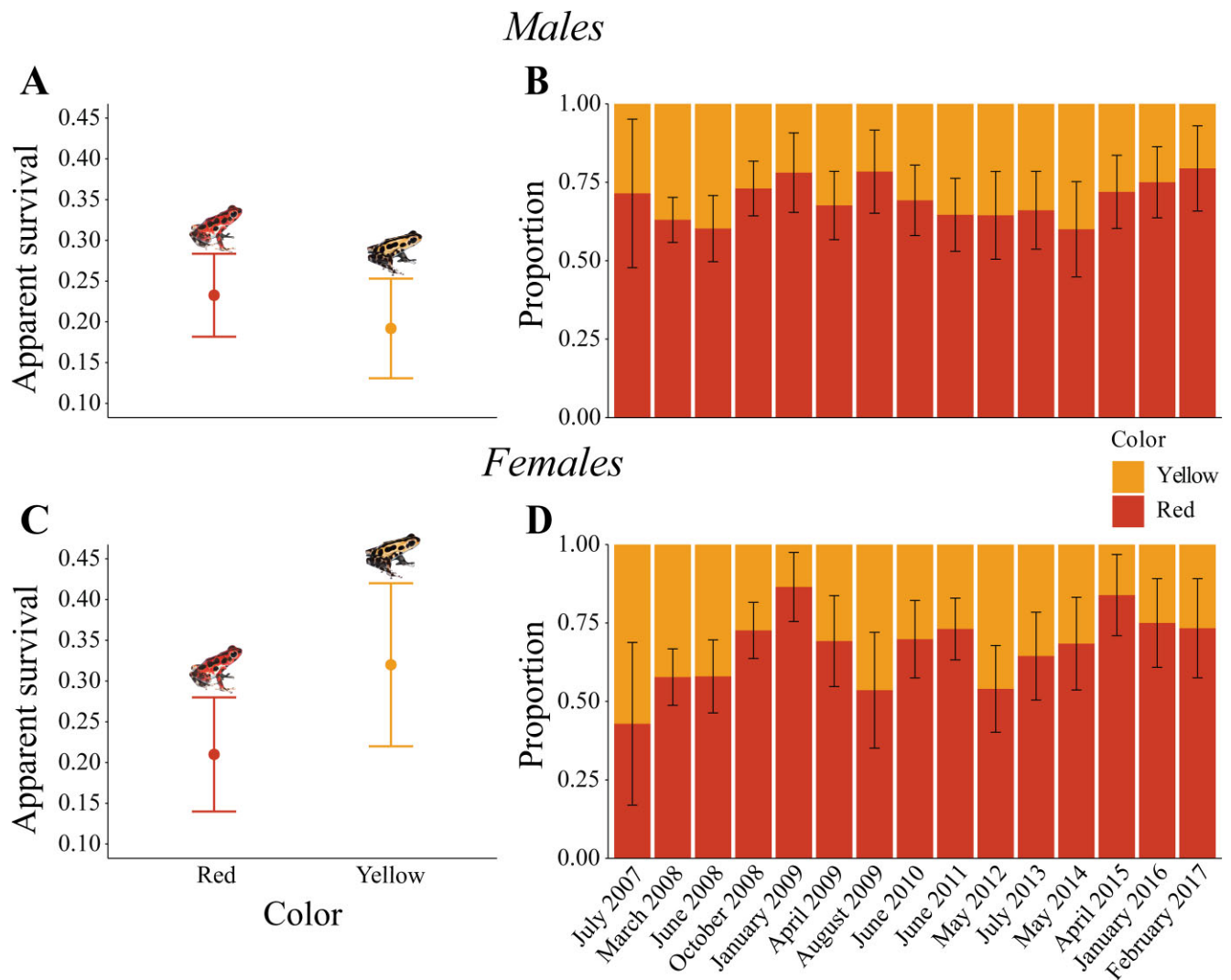


Figure 2. (A) and (C) represent the apparent survival estimates of red and yellow male and female frogs for the entire study period. (A) is based on estimates of the second-best model for males (survival dependent on color but detection constant) while (B) is based on estimates of the best model for females (survival dependent on color but detection constant). The dots at the middle of the bar represent the estimated mean values and the bars represent 95% confidence intervals. (B) and (D) represent the proportions of red and yellow male and female frogs captured in the study area for every survey period. Black bars represent 95% confidence intervals.

mation criterion, QAICc, in *RMark*). Using this method, the best model has a ΔQAICc value of zero and models with a $\Delta\text{QAICc} > 2$ are considered to have less support. We then performed likelihood ratio tests for nested models with a $\Delta\text{QAICc} < 2$ to test the null hypothesis that the reduced model is a better fit to the data than models with additional parameters.

Simulated territorial intrusion assays

We conducted simulated territorial intrusion assays at the same study site in July–August of 2015 and February–March of 2016, following protocols in Yang et al. (2018). Briefly, we used 3D printed and hand painted plastic decoy frogs, coupled with decoy movement and call playback, to simulate territory intrusions on territorial male *O. pumilio* in the field. We painted the frog decoys either red or yellow with acrylic paints and used a spectrometer to ensure that the color of the decoy matched the reflectance spectrum of frogs of each color morph from the study popula-

tion. The frog decoys were mounted on remote-controlled servomotors hidden within wood-textured, log-shaped containers (which serve as the decoy's perch), and Bluetooth speakers were attached to the containers for advertisement call playback (see Yang et al., 2018 for additional details).

We conducted the assays between 700 and 1,300 hr when *O. pumilio* is most active (Graves, 1999; Graves et al., 2005). Upon locating a calling, territorial male, we placed the decoy apparatus within a meter of his perch, then allowed the male to acclimate to the disturbance for 5 min or until when he started calling again. The color of the frog decoy was chosen randomly for each trial. We began the assay by starting the call playback and moving the decoy frog using the remote control and recorded the territory defense behaviors the focal male displayed in response. We noted the first occurrence and the latency to perform each of four (typically sequential) behaviors that represent increasing levels of aggression (Yang et al., 2018): (i) “track”: turning to face the frog decoy, (ii) “approach”: moving closer to the frog decoy,

(iii) “challenge”: continuing the approach onto the frog decoy’s perch, and (iv) “attack”: physically pushing, kicking, or wrestling the frog decoy. We also recorded the total number of aggressive call bouts in a trial. We noted the start of a trial when the focal male first displayed an aggressive intention (i.e., track). For males that attacked the decoy, we extended or shortened each trial to allow a consistent 5 min of interaction after the first attack, and tallied the number of times the male attacked the frog decoy within those 5 min to quantify the intensiveness of an escalated contest. Trials can therefore range from 5 (immediate attack) to 15 min (attack at min 10) after the focal male tracked the frog decoy. If the focal male failed to track the frog decoy within 15 min, we terminated the trial and discarded the observation as the lack of response could have been due to the apparatus being placed outside of the male’s territory.

After each assay, we captured the male and took its photographs and measurements using the same equipment and protocol as for the capture–mark–recapture study described above. Males were only assayed once, and we avoided resampling individuals by marking each tested male with a toe clip (Funk et al., 2005). We also measured perch height (distance from the ground where the male was observed calling prior to the trial) and recorded whether a conspecific frog was nearby during the trial (yes/no) to include these as covariates in the analyses.

Statistical analysis for simulated territorial intrusion assays

To test whether the level of male territorial aggression was influenced by color, we examined the effects of male color (red/yellow), decoy color (red/yellow), and their interaction on the latency for a territorial male to (i) track, (ii) approach, (iii) challenge, and (iv) attack the frog model using Cox proportional hazards regressions. Using the same predictor variables, we also modeled whether (yes/no) a territorial male (i) challenged or (ii) attacked the frog decoy using binomial generalized linear models (GLMs). We did not perform analyses on whether or not a male performed track, approach, or call behaviors because > 90% of males performed these behaviors, and this outcome imbalance made model estimations unreliable. In the subset of males that called during the trial, we used a negative binomial GLM to test the effect of male color morph, decoy color, and their interaction on the number of call bouts, including total trial time as a statistical offset to account for differences in trial length. In the subset of males that attacked the decoy, we used a negative binomial GLM to test the effect of male color morph, decoy color, and their interaction on the number of attacks during the 5 min following the first attack.

For all statistical models described above, we included perch height and conspecific presence (yes/no) as covariates. Nonsignificant covariates and interaction terms were examined and dropped based on AIC. When the male color morph $\times$ decoy color interaction was significant, we performed a post hoc analysis (with multivariate t multiple comparison correction) to examine whether red and yellow males show different responses toward the two decoy colors.

All statistical analyses were performed in R version 3.3.2 (R Core Team, 2022). The Cox proportional hazards regressions were performed using the *coxph* function in the *sur-*

vival package (Therneau, 2022). The binomial GLMs were performed using the *glm* function in the *stats* package (R Core Team, 2022), and the negative binomial GLMs were performed using the *glm.nb* function in the *MASS* package (Venables & Ripley, 2002). AIC model selections were performed using the *dredge* function in the *MuMIn* package (Barton, 2022), and the significance of the terms that remained in the final models were calculated using a likelihood ratio test using the *Anova* function in the *car* package (Fox et al., 2022). Post hoc comparisons were performed using the *emmeans* function in the package *emmeans* (Lenth, 2022).

Results

Modeling the effect of color morph on survival

Males

Across the 15 surveys, we captured a total of 706 males (229 yellow and 477 red) of which 231 were recaptured at least once (70 yellow and 161 red). The proportions of red and yellow males remained fairly constant across surveys, and the red morph (average proportion across sampling periods $\pm$ average 95% CI across sampling periods: 0.69 ± 0.13) was consistently more frequent than the yellow morph (0.31 ± 0.13 ; Figure 2B).

In our CJS exercise, the goodness of fit test did not show a significant lack of fit of the built models to our data. We corrected the AIC value of every model using the $\hat{c}$ value obtained for the full model (4.36 with sampling standard error = 0.81, indicating overdispersion) to obtain QAICc values. Out of the 16 models we tried, three of them fit the data similarly ($\Delta\text{QAICc} < 2$): survival and detection constant; survival dependent on color but detection constant; survival constant but detection dependent on color (Table 1). Because the simplest model (survival and detection constant) is nested within the other two models, we performed likelihood ratio tests between the simplest model and the other two more complex models. Neither test resulted in a *p*-value less than 0.05 (survival and detection constant vs survival dependent on color but detection constant: *p* = 0.32; survival and detection constant vs survival constant but detection dependent on color: *p* = 0.49), suggesting that the simplest model (survival and detection constant) is the best fit to our data. Therefore, we interpreted the probability of survival and detection as being equal between males of both color morphs, with the average apparent survival estimate (i.e., the average probability of one male frog surviving to the next survey occasion) being (apparent survival $\pm$ 95% CI: 0.22 ± 0.04). Our interpretation of similar survival among morphs for males is further supported by the fact that even in the less supported model in which survival is dependent on color (Table 1, second best model), the average apparent survival estimates overlap greatly between morphs (apparent survival $\pm$ 95% CI: red = 0.23 ± 0.05 ; yellow = 0.19 ± 0.06) (Figure 2A).

Females

We captured a total of 645 females (206 yellow and 439 red) of which 114 were recaptured at least once (46 yellow and 68 red). Except for one sampling occasion (July 2007), the proportions of red and yellow female frogs remained fairly constant across surveys, with red (average proportion across

Table 1. Summary results of CJS model ranking for the best models for males and females. Models are organized from the best supported model ($\Delta\text{QAICc} = 0$) to the least supported model. K = number of parameters; QAICc = Akaike's small-sample-size-corrected quasi-likelihood information criterion, and ΔQAICc = Delta of the QAICc between every model and the best model. Bolded values indicate the two models that fit best our data for males and females respectively. The rest of the models reported from *RMark* are in Table S1.

Model		K	QAICc	ΔQAICc	Weight	Deviance
<i>Males</i>						
Survival	Detection					
Constant	Constant	2	314.1361	0.0000	0.5100	65.5213
Color	Constant	3	315.9248	1.7887	0.2085	65.2966
Constant	Color	3	316.0413	1.9052	0.1967	65.4131
<i>Females</i>						
Survival	Detection					
Color	Constant	3	493.4937	0.0000	0.3474	111.1284
Constant	Constant	2	493.8840	0.3904	0.2858	113.5354
Constant	Color	3	494.9900	1.4964	0.1644	112.6248
Color	Color	4	495.4595	1.9658	0.1300	111.0721

Table 2. Cox proportional hazards models and GLMs testing the effects of male color morph, decoy color, and their interaction on selected parameters of aggression in simulated territorial intrusions (see Table S2 for results of unrepresented parameters). (i) Latency to attack the decoy, (ii) likelihood to attack the decoy, (iii) number of aggressive call bouts during the trial, and (iv) number of attacks in 5 min after first attack. Perch height and conspecific presence were included as covariates. The "-" symbol indicates that the interaction or covariate was dropped from the model for a particular response variable. Bolded values indicate models, where the predictor was found to have statistical significance.

Predictors	df	(i) Attack latency			(ii) Attack (Y/N)			(iii) # of Calls			(iv) # of Attacks		
		β	SE	<i>p</i>	β	SE	<i>p</i>	β	SE	<i>p</i>	β	SE	<i>p</i>
Male color ¹	1	-0.17	0.34	0.641	-0.74	0.65	0.250	-0.19	0.16	0.241	0.13	0.21	0.538
Decoy color ²	1	-1.05	0.32	0.013	-1.91	0.65	0.002	-0.46	0.16	0.004	-0.11	0.21	0.584
Male ¹ × decoy ²	1	1.20	0.60	0.042	1.99	0.92	0.027	0.40	0.23	0.090	-	-	-
Perch height	1	0.00	0.00	0.103	-	-	-	-	-	-	-	-	-
Conspecific ³	1	-	-	-	-	-	-	-0.28	0.17	0.097	-	-	-

¹baseline = red; ²baseline = red; ³baseline = no conspecifics.

sampling periods $\pm$ average 95% CI across sampling periods: 0.67 ± 0.14) being more common than yellow frogs (0.33 ± 0.14 ; Figure 2D).

In our CJS exercise, the goodness of fit test did not show a significant lack of fit of the built models to our data. We corrected the AIC value of every model using the $\hat{c}$ value obtained for the most general model (1.71 with sampling standard error = 0.38, indicating overdispersion) to obtain QAICc values. Out of the 16 models we compared, four of them fit the data similarly ($\Delta\text{QAICc} < 2$): survival dependent on color but detection constant; survival and detection constant; survival constant and detection dependent on color; and both survival and detection dependent on color (Table 1). Because the simplest model (survival and detection constant) is nested within the other three models, we performed likelihood ratio tests between the simplest model and the other three more complex models. The result was not significant when we compared the simplest model with the third or fourth best models (survival and detection constant vs both survival and detection dependent on color: $p = 0.12$; survival and detection constant vs. survival constant but detection dependent on color: $p = 0.21$). However, when we compared the simplest model and the best model, the p -value was less than 0.05 (survival and detection constant vs. survival dependent on color but detection constant: $p = 0.04$), suggesting that the more complex model (survival dependent on color but detection constant) is the best fit for our female data. We interpreted this as evidence that the probability of survival differs between color morphs for females. In this model, apparent survival (i.e., the average probability of one frog surviving to the next survey period, shown as:

apparent survival $\pm$ 95% CI) is higher for yellow females (0.32 ± 0.10) than for red females (0.21 ± 0.07) (Figure 2C).

Simulated territorial intrusion assays

We collected data on a total of 88 successful assays: 47 on red males (25 with a red decoy, 22 with a yellow decoy) and 41 on yellow males (20 with a red decoy, 21 with a yellow decoy). Of these males, 92% approached, 90% called at, 80% challenged, and 58% attacked the frog decoy.

Male color and decoy color did not influence the latency for males to track, approach, or challenge (Table S2) but did interact to influence their latency to attack (Table 2). Red males attacked the decoy faster when the decoy was red than when it was yellow ($\beta \pm \text{SE} = 1.05 \pm 0.45$, $z = 2.35$, $p = 0.037$), but attack latency for yellow males did not depend on decoy color ($\beta \pm \text{SE} = -0.15 \pm 0.40$, $z = -0.361$, $p = 0.920$; Figure 3A). Male color and decoy color also interacted to influence the likelihood that a male would challenge (Table S2) or attack the decoy (Table 2), but post hoc tests found no significant difference in challenge likelihood toward red vs. yellow decoys for either red males ($\beta \pm \text{SE} = 2.19 \pm 1.13$, $z = 1.95$, $p = 0.100$) or yellow males ($\beta \pm \text{SE} = -0.316 \pm 0.708$, $z = -0.446$, $p = 0.881$). Red males were 5.8 times more likely to attack a decoy when the decoy was red than when it was yellow ($\beta \pm \text{SE} = 1.91 \pm 0.66$, $z = 2.92$, $p = 0.007$), but yellow males were equally likely to attack a red and a yellow decoy ($\beta \pm \text{SE} = -0.08 \pm 0.64$, $z = -0.13$, $p = 0.990$; Figure 3B). We found a marginal male color $\times$ decoy color interaction on number of aggressive call

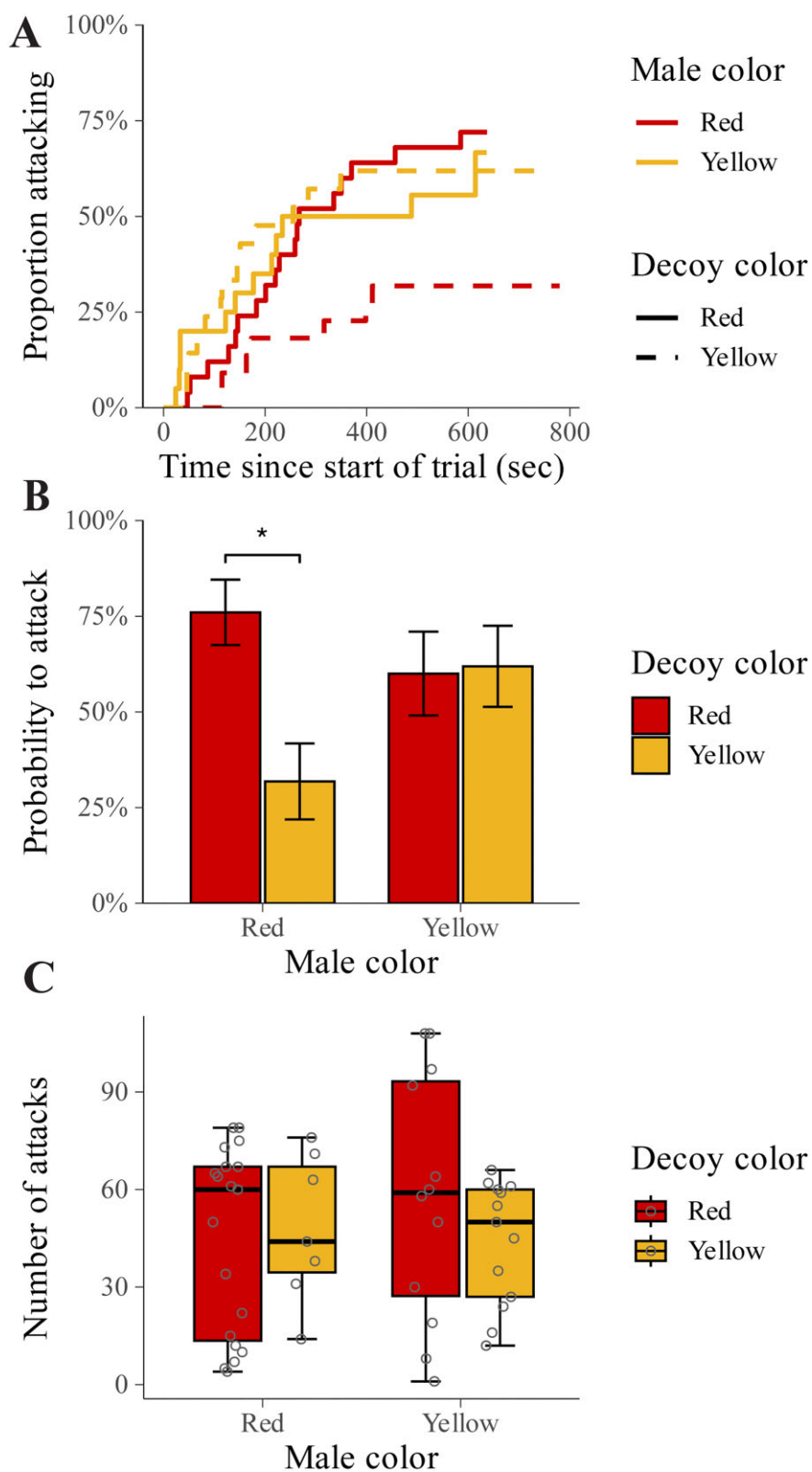


Figure 3. (A) Proportion and latency of attacks. (B) Probability of a male attacking the decoy. Error bars are standard errors estimated from binomial GLMs. (C) Number of attacks on the decoy following the first attack. Bold lines indicate medians, boxes enclose 25th to 75th percentiles, and error bars enclose data range.

bouts (Table 2), and post hoc tests revealed that red males called more when the decoy was red than when it was yellow ($\beta \pm SE = 0.46 \pm 0.16$, $z = 2.91$, $p = 0.007$), but yellow males called equally frequently toward a red and a yellow decoy ($\beta \pm SE = 0.06 \pm 0.17$, $z = 0.45$, $p = 0.929$). Of the males that attacked the decoy, neither male color nor decoy color influenced the number of attacks in 5 min (Table 2; Figure 3C).

Discussion

In our study, we investigated the roles of natural selection and both inter- and intrasexual selection on the maintenance of polymorphism in an aposematic signal. Predation is hypothesized to generate positive frequency dependent selection that tend to destabilize polymorphisms in aposematic signals. Intersexual selection that leads to assortative mating among males and females with similar warning signals should also destabilize such polymorphisms by causing an increase in frequency and eventual fixation of the more common morph. On the other hand, if the aposematic signal has been co-opted to function in intrasexual competition, this could lead to reduced competition and more access to mates for the rarer morph (negative frequency dependent selection), a mechanism of sexual selection that could act to stabilize an aposematic polymorphism.

To investigate this idea, we tested for differences in survival and intrasexual competition among two sympatric aposematic color morphs of the strawberry poison frog, *O. pumilio*. Because our models did not explicitly test for differences among the sexes, we focus here on differences among color morphs within each sex. Our 10-year capture mark recapture study revealed that females of the rarer yellow morph show higher survival than red females, while red and yellow males show similar survival. Using simulated territorial intrusion assays, we also showed that red territorial males were more aggressive toward red male intruders compared to yellow intruders. However, yellow males are equally aggressive toward both red and yellow intruders. Below we discuss the implications of these findings, in the context of previous work, for our understanding the evolution of aposematic signal polymorphisms in nature.

Our capture-mark-recapture study revealed that males of the more common red *O. pumilio* morph had similar survival rates compared to males of the rarer yellow morph. The two morphs are easily distinguishable in a variety of potential predator visual systems (e.g., human, bird, and crab; Richards-Zawacki et al., 2013) but are equally unpalatable based on their skin chemical profiles (Saporito et al., 2006). Contrary to expectations from aposematism theory, red males, which are more common locally and regionally, do not appear to gain a survival benefit compared to the rarer yellow males. This result is consistent with previous work using clay models that shows similar attack rates on both red and yellow morphs from this population (Richards-Zawacki et al., 2013). Two alternative mechanisms could explain our observations of predation risk for both red and yellow frogs: signal generalization, and equal effectiveness of the signals. Some predators may learn from one warning signal and generalize their learned avoidance to other phenotypes they perceive as similar. For instance, Amézquita et al. (2013) found that predators generalize learned avoidance to a wide range of color morphs (from red, orange, to yellow)

present in the polytypic congener *Oophaga histrionica*. Alternatively, multiple distinct but coexisting warning signals may be learned simultaneously and effectively by predators, regardless of their relative frequencies in the population. For instance, young domestic chickens readily learn to avoid unpalatable *O. pumilio* and their rate of their learning is not influenced by whether they experienced only one morph or three different color morphs during training (Yeager, 2015). Finally, the red and yellow morphs of *O. pumilio* may represent equally effective, general warning signals to visual predators. Red and yellow are typical warning colors that have evolved many times in aposematic prey spanning numerous taxa (Ham et al., 2006; Lindström, 1999) (Figure 1), leading to widespread innate and learned predator avoidance of these colors (Marples et al., 1998; Skelhorn, 2011). If red and yellow are in fact equally effective warning signals for *O. pumilio*, this would suggest that while predation pressure failed to generate the predicted survival advantage for the more common morph, it may have constrained the evolution of phenotypes in this population to bright colors that are effective as warning signals.

If both aposematic signals are equally effective in deterring predators, the question remaining is: why do red and yellow females have different survival rates between them? We hypothesized that this result may be explained by differences in sexual behaviors among the color morphs, rather than the predation risks they experience. *Oophaga pumilio* has a promiscuous mating system in which males defend territories, and females choose mates from among the males whose territories overlap the female's home range (Pröhl & Berke, 2001). Both red and yellow females show assortative preferences for their own color when assayed in laboratory conditions (Richards-Zawacki & Cummings, 2011). However, red females in the wild mate assortatively (i.e., more often with red males than expected by chance) while yellow females mate indiscriminately with respect to male color (Richards-Zawacki et al., 2012). This suggests that red females are choosier than yellow females, an asymmetric behavioral bias that mirrors what we saw for male aggression patterns among the two morphs in this study. For females, being choosy often comes at a cost, including increased risk of predation, increased energy expenditure, and/or less reproduction if males of the desired phenotype are difficult to find (Real, 1990; Yukilevich & Aoki, 2022). Since yellow is the rarer morph in our study population, it stands to reason that for yellow females, who would encounter yellow males less easily than red males, choosiness could be more costly. Choosiness may be especially costly in species with extended courtship or mating, like *O. pumilio*, where individuals spend long periods of time exposed to predators and do little to no foraging while engaging in these activities. *Oophaga pumilio* females court for hours before choosing to deposit their eggs and they often abandon courtship, even after hours of engagement (Yang, Blomenkamp et al., 2019). Our capture-mark-recapture study found higher survival rate for the less choosy yellow females, who presumably spend less time active and exposed during courtship than red females and males of both morphs (who are exposed when engaged in both courtship and territory defense). This suggests that there may be a fitness trade-off between mate choosiness and survival. Differences in mate choosiness are often seen between sexes but rarely examined among distinct phenotypes of the same sex (Courtiol et al., 2016; Jennions

& Petrie, 2007). An exception is the alternative male mating tactics associated with different levels of choosiness in male side-blotched lizards, which are thought to be generated by oscillations in male morph frequencies (Sinervo & Lively, 1996). The optimal level of choosiness should depend on the costs of choosiness; as costs go up, individuals should evolve to be less choosy (Yukilevich & Aoki, 2022). It is possible that one mechanism for the reduction of choosiness in a population is reduced survival of the choosier individuals, as seen in our study.

While survival estimates from mark–recapture suggest that predation pressure for red and yellow males is similar, behavioral assay data suggests that red and yellow males are under different sexual selection pressure. The color-assortative mate choice manifested by the more common red females in the wild (Richards-Zawacki et al., 2012) is expected to exert positive frequency dependent selection that favors males of the common red morph. Over time, this should lead to the increase and eventual fixation of the red morph. However, intrasexual selection may act in opposition, favoring males of the rare morph and stabilizing the polymorphism. Our territorial intrusion experiment showed that red males bias their aggression toward red rivals, but yellow males are similarly aggressive to both phenotypes. Because of this asymmetric aggression bias, yellow males should enjoy a fitness advantage due to reduced aggressive interactions during territorial disputes. Yellow males may thus have an easier time establishing a territory and consequently may hold more territories than red males in proportion to their frequency in the population. The aggression a territorial male receives can also affect his reproductive success even when territories do not change hands, for example if spending more time defending territories from intruders means more time out from courtship and mating (Yang & Richards-Zawacki, 2021). Intrasexual selection is recognized as an important generator of diversity in male and female traits across animal taxa (e.g., birds, Cally et al., 2021; cichlid fish, Dijkstra et al., 2007; and lizards, Pérez I De Lanuza et al., 2017) but has only recently begun to be explored in aposematic species (Yang, Servedio et al., 2019). This is, perhaps, due to the assumption that aposematic traits are primarily shaped by natural selection, and that sexual selection is primarily driven by mate choice rather than competition for mating opportunities among members of the same sex.

An interesting question brought about by this work is whether the observed effects of predation, intraspecific competition, and mate choice would be sufficient to maintain the polymorphism. To answer this question, simulation models could be used in which the effect of these variables on the fate of this polymorphism (stable coexistence vs. fixation of a single morph) could be observed over time. Moreover, we suggest that future research on the maintenance of polymorphisms should not discard alternative mechanisms, such as sexual conflict, which might lead to rapid evolutionary change in morph frequencies (Bonduriansky, 2011; Svensson et al., 2005) but does not necessarily apply in our system. For example, harassment from *Heliconius* males toward mated females is theorized to contribute to local polymorphisms by favoring less-preferred morphs that do not suffer harassment, yet empirical evidence is still required (Hausmann et al., 2023).

Our study underscores the importance of considering the joint action of multiple sources of selection to better understand the evolution of aposematic signals. Consideration of both inter- and intrasexual selection, in addition to natural selection, may be needed to resolve the classic evolutionary paradox of the aposematic polymorphism.

Supplementary material

Supplementary material is available online at *Evolution*.

Data availability

All data and code used in the manuscript has been archived on Dryad. The archives can be found on <https://doi.org/10.5061/dryad.hhmgqknkvf>

Author contributions

M.G-S., Y.Y., and C.L.R-Z. were involved in the conceptualization of the study. All authors were involved in data collection. M.G-S., R.P.R., and Y.Y. were responsible for the data curation. The formal analysis and investigation were conducted by M.G-S., Y.Y., and C.L.R-Z. C.L.R-Z acquired funding for this project. Resources and supervision were provided by C.L.R-Z. M.G-S., Y.Y., and C.L.R-Z wrote the original draft.

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Conflict of interest

The authors declare no conflict of interests.

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